

Ansible Lab: Ad-hoc Commands & Playbooks

Part 1: Ad-hoc Commands (5 Questions)

- 1. Use an ad-hoc command to check if all hosts are reachable using the ping module.

```
abdelhamed@server1:~/Desktop/Ansible$ ansible all -i inventory -m ping
localhost | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
```

- 2. Write an ad-hoc command to create a directory named /tmp/testdir on all managed nodes.

```
abdelhamed@server1:~/Desktop/Ansible$ ansible all -i inventory -m file -a 'path=/tmp/testdir state=touch'
localhost | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "dest": "/tmp/testdir",
  "gid": 1000,
  "group": "abdelhamed",
  "mode": "0644",
  "owner": "abdelhamed",
  "secontext": "unconfined_u:object_r:user_tmp_t:s0",
  "size": 0,
  "state": "file",
  "uid": 1000
}
```

- 3. Use an ad-hoc command to install the 'git' package using the yum module.

```
root@server1:/home/abdelhamed/Desktop/Ansible# ansible all -i inventory -m yum -a 'name=git state=latest'
localhost | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "msg": "Nothing to do",
  "rc": 0,
  "results": []
}
```

- 4. Write an ad-hoc command to copy a local file named 'index.html' to '/tmp/index.html' on all web servers.

```
root@server1:/home/abdelhamed/Desktop/Ansible# ansible all -i inventory -m copy -a 'src=/usr/share/www/index.html dest=/tmp/index.html'
localhost | SUCCESS => {
  "changed": false,
  "dest": "/tmp/index.html",
  "src": "/usr/share/www/index.html"
}
```

- 5. Use an ad-hoc command to restart the 'nginx' service on all hosts that have it installed.

```
root@server1:/home/abdelhamed/Desktop/Ansible# ansible all -i inventory -m service -a 'name=nginx state=restarted'
localhost | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "name": "nginx",
}
```

◇ 8. Change File Permissions

- `ansible all -m file -a "path=/tmp/file.txt mode=0644"`
 -
 - `ansible all -m setup` → For the ansible facts
 - Filter specific facts:
 - `ansible all -m setup -a "filter = ansible_distribution"` → the os family type [Debian – redhat - fedora]
-

◇ 10. Delete a File

- `ansible all -m file -a "path=/tmp/file.txt state=absent"`
 - Your Turn please try all this commands:
 -
 - Create a file
 - `ansible all -m file -a "path=/tmp/demo.txt state=touch"`
 - ◇ Delete a directory
 - `ansible all -m file -a "path=/tmp/testdir state=absent"`
-

 USER & GROUP MANAGEMENT

- ◇ Create a group
 - `ansible all -m group -a "name=devops state=present" -b` → -b for become sudo0
 - ◇ Add a user to a group
 - `ansible all -m user -a "name=devops groups=wheel append=yes" -b`
-

 PACKAGE MANAGEMENT

- ◇ Remove a package
 - `ansible all -m yum -a "name=httpd state=absent" -b`
 - ◇ Upgrade all packages
 - `ansible all -m yum -a "name=* state=latest" -b`
-

 SERVICE MANAGEMENT

- ◇ Stop a service
 - `ansible all -m service -a "name=nginx state=stopped" -b`
 - ◇ Enable a service on boot
 - `ansible all -m service -a "name=nginx enabled=yes" -b`
-

 FILE CONTENT OPERATIONS

- ◇ Append a line to a file
 - `bash`
 - `CopyEdit`
 - `ansible all -m lineinfile -a "path=/etc/motd line='Welcome to the server!'" -b`
 - ◇ Replace a line in a file (VI)
 - `ansible all -m lineinfile -a "path=/etc/ssh/sshd_config regexp='^PermitRootLogin' line='PermitRootLogin no'" -b`
-

 PERMISSION & OWNERSHIP

- ◇ Change file owner
 - `ansible all -m file -a "path=/tmp/demo.txt owner=ec2-user group=ec2-user" -b`
-

-  FETCH & PUSH FILES
- ◇ Fetch a file from remote nodes to control node but fetch the destination must be a directory as it will make the tree of the files
- Flat yes -> will make the the last host as file
- Flat no -> will make the default behaviour for the fetch command where it can be like a tree
- `ansible all -m fetch -a "src=/etc/hosts dest=./hosts_backup flat=yes"`

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-  GATHER FACTS
 - ◇ Show IP address of each node
 - `ansible all -m setup -a "filter=ansible_default_ipv4.address"`

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-  LOOP-LIKE BEHAVIOR (with shell)
 - ◇ List all `.log` files in `/var/log`
 - `ansible all -m shell -a "ls /var/log/*.log"`
 - ◇ Run a custom shell script
 - `ansible all -m script -a "./deploy.sh"`

- `#!/bin/bash`
- `sudo yum install -y httpd`
- `sudo systemctl start httpd`
- `sudo systemctl enable httpd`